Figure 1. Study design and BBJ discovery

A Blood expression instruments
4,462 druggable genes considered
2,544 genes; 6,135 SNP–gene records

→

BBJ GD discovery
2,234 estimable genes; 13 discoveries
Threshold P < 0.05/2,544

→

Disease-definition comparisons
FinnGen GD, UKB hyperthyroidism
Two overlapping GO definitions

Regional follow-up: TSHR and IGF1R at the centre; CTLA4 comparator; 24 selected genes in total.

B BBJ gene-level MR associations

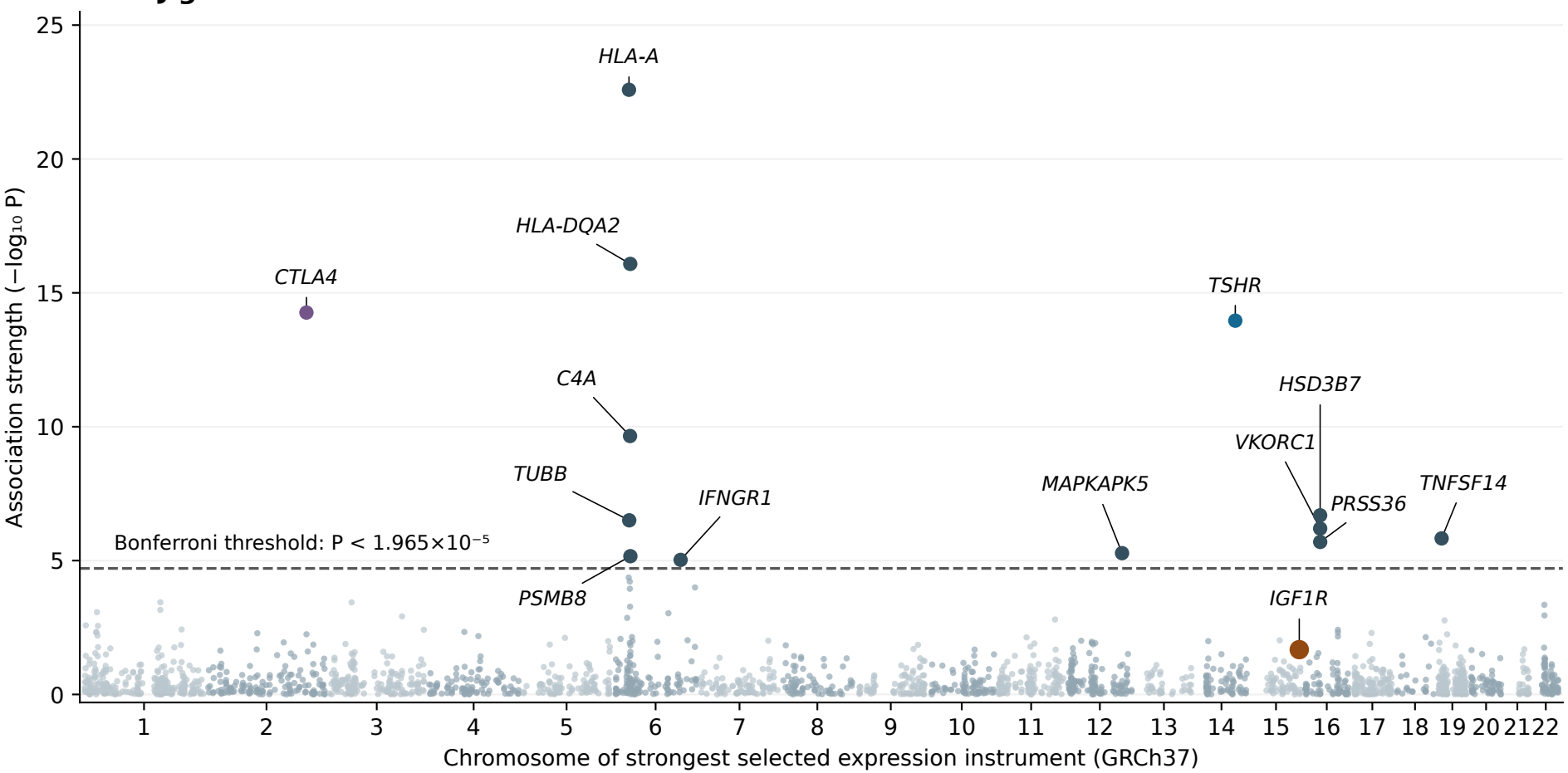


Figure 2. TSHR and IGF1R across disease definitions, with CTLA4 as comparator

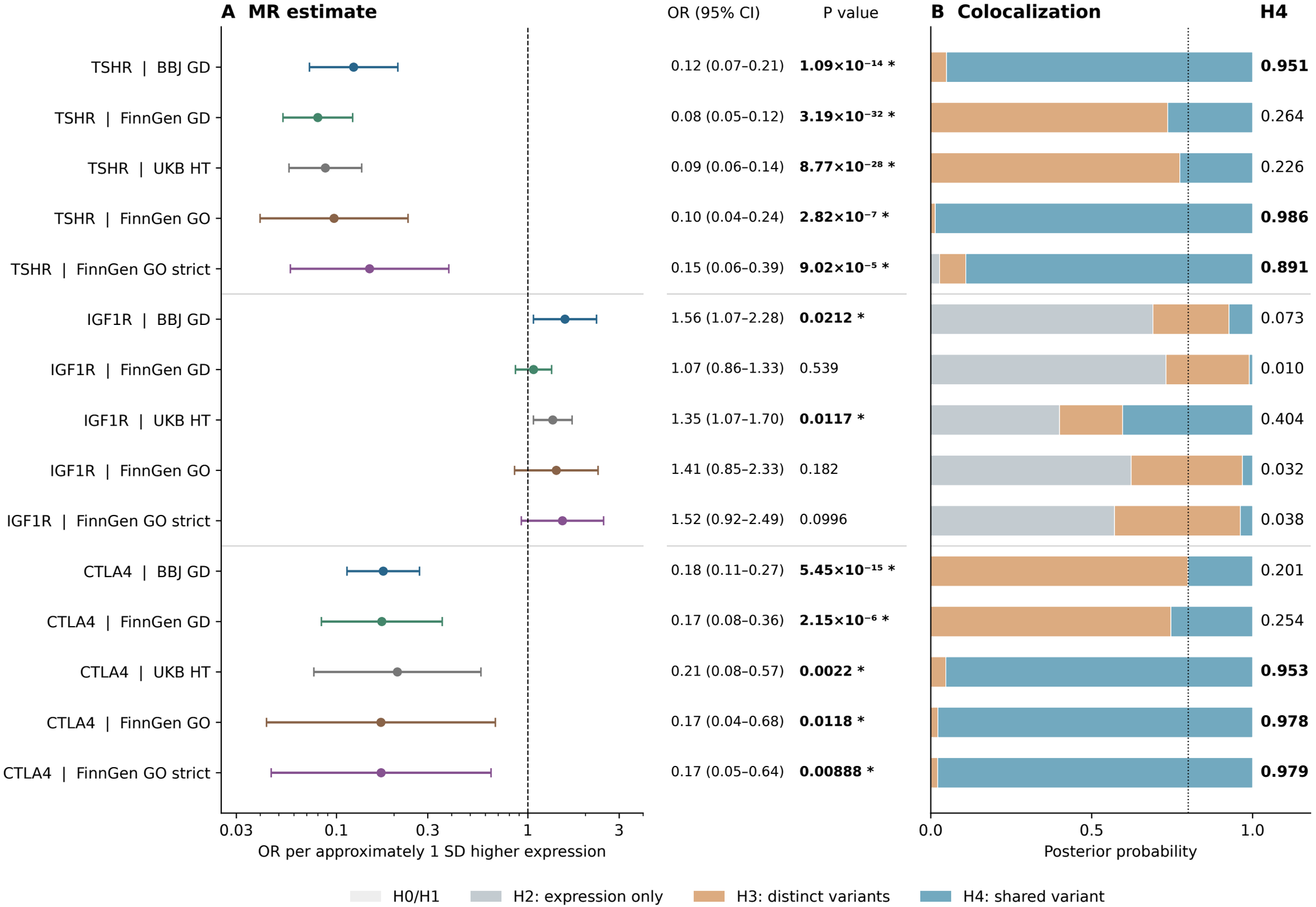
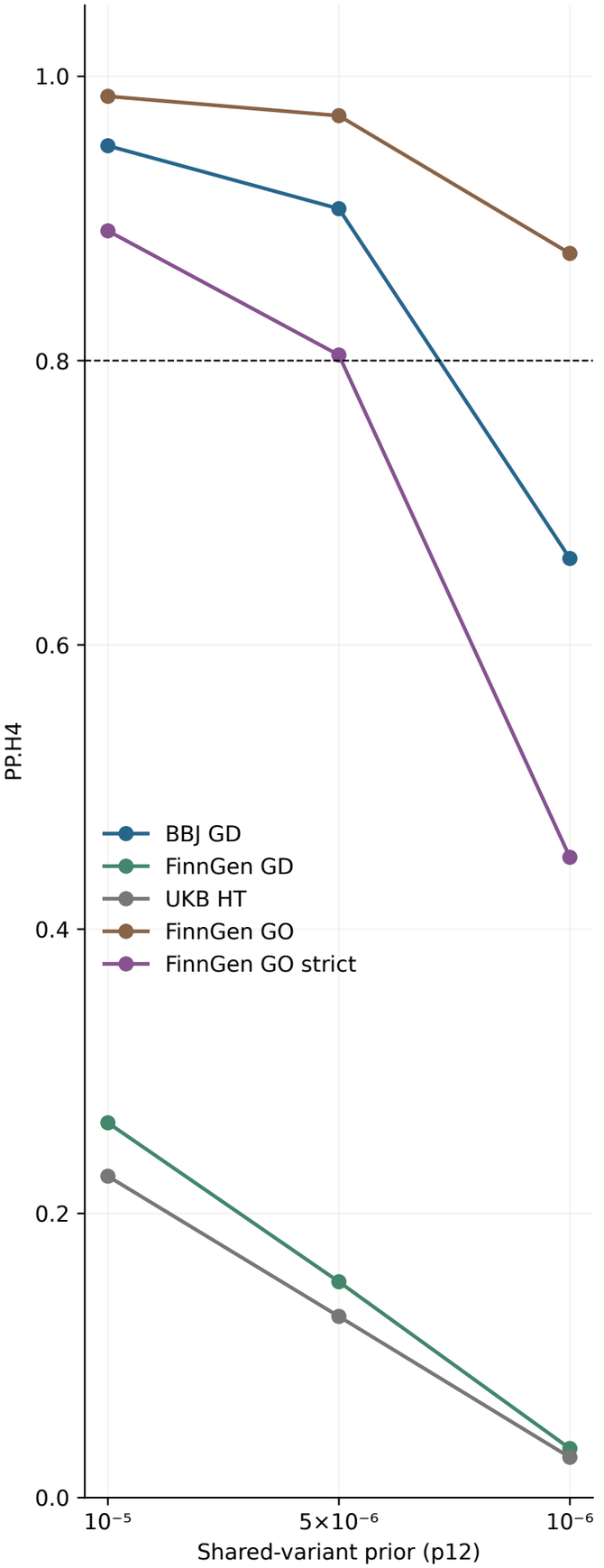


Figure 3. Regional genetic evidence and TSHR prior sensitivity

A Shared-variant probability (PP.H4)

	BBJ GD	FinnGen GD	UKB HT	FinnGen GO	FinnGen GO strict
TSHR	0.951	0.264	0.226	0.986	0.891
IGF1R	0.073	0.010	0.404	0.032	0.038
CTLA4	0.201	0.254	0.953	0.978	0.979
HLA-A	<0.001	<0.001	<0.001	<0.001	<0.001
HLA-DQA2	<0.001	<0.001	<0.001	<0.001	<0.001
C4A	<0.001	<0.001	<0.001	0.030	0.326
HSD3B7	0.636	0.011	0.095	0.026	0.046
TUBB	<0.001	<0.001	<0.001	<0.001	<0.001
VKORC1	0.388	0.011	0.030	0.030	0.048
TNFSF14	0.994	0.303	0.237	0.017	0.018
PRSS36	0.168	0.063	0.115	0.037	0.033
MAPKAPK5	<0.001	0.041	0.005	0.026	0.029
PSMB8	<0.001	<0.001	<0.001	<0.001	<0.001
IFNGR1	0.989	0.313	0.052	0.020	0.021
APOM	<0.001	<0.001	<0.001	0.006	0.227
NEU1	<0.001	<0.001	<0.001	0.003	0.061
IGLV2-18	0.029	0.020	0.024	0.082	0.100
IGLV3-12	0.019	0.022	0.024	0.083	0.101
TNF	<0.001	<0.001	<0.001	<0.001	<0.001
HIST1H4E	<0.001	<0.001	<0.001	0.004	0.007
LTA	<0.001	<0.001	<0.001	<0.001	<0.001
LTB	<0.001	<0.001	<0.001	<0.001	<0.001
C4B	<0.001	<0.001	<0.001	0.030	0.311
HLA-G	<0.001	<0.001	<0.001	<0.001	0.001

B TSHR prior sensitivity



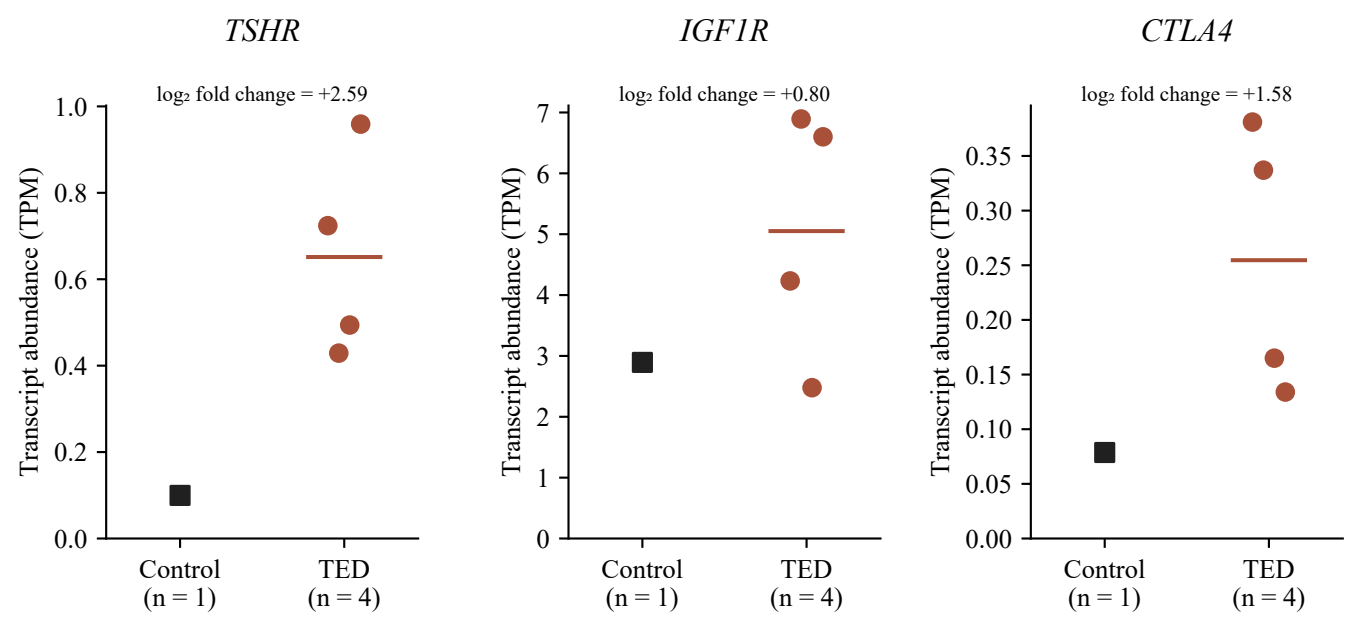
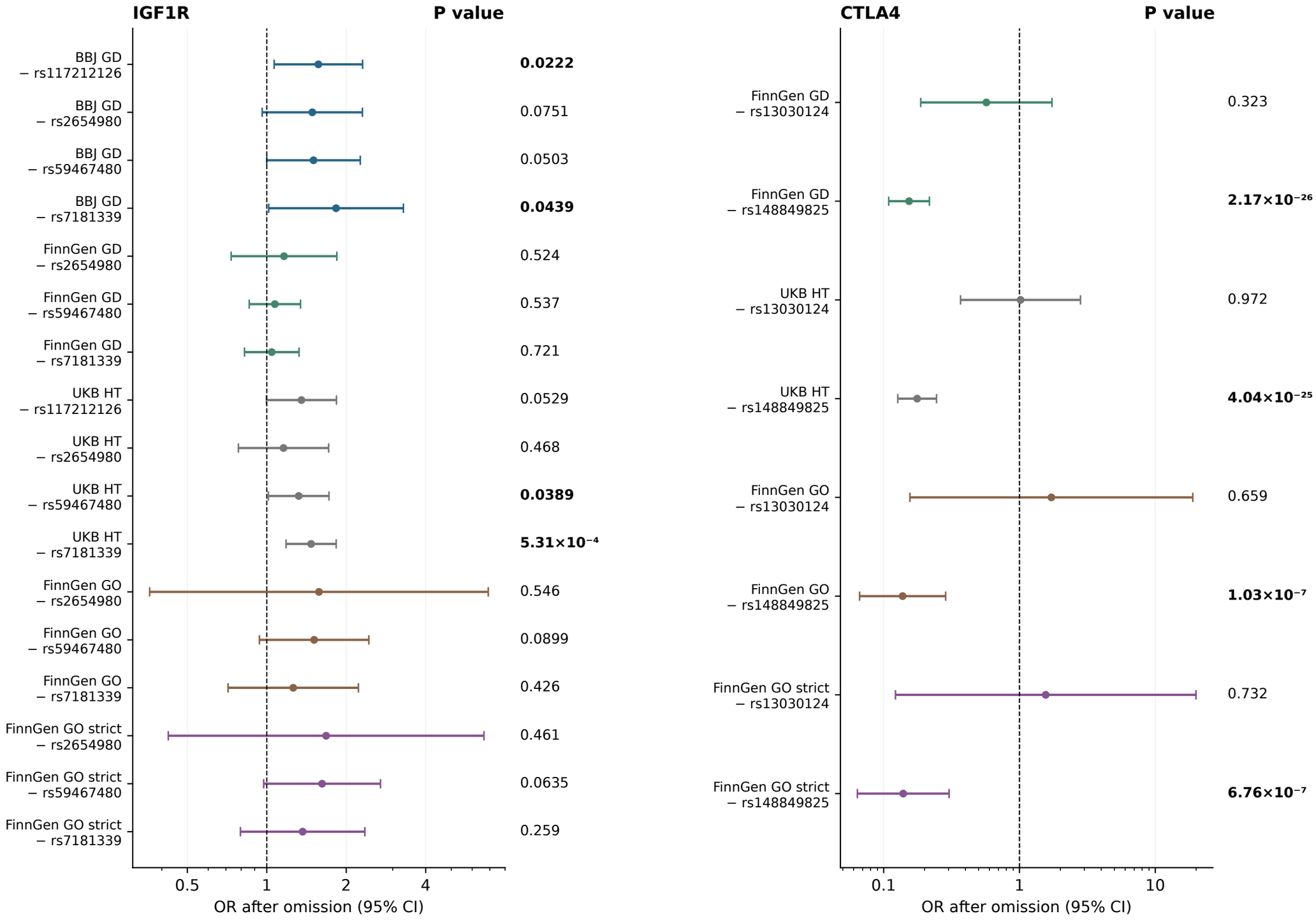

TSHR
log₂ fold change = +2.59
Transcript abundance (TPM)
1.0
0.8
0.6
0.4
0.2
0.0
Control
(n = 1)
TED
(n = 4)
IGF1R
log₂ fold change = +0.80
Transcript abundance (TPM)
7
6
5
4
3
2
1
0
Control
(n = 1)
TED
(n = 4)
CTLA4
log₂ fold change = +1.58
Transcript abundance (TPM)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Control
(n = 1)
TED
(n = 4)

Figure S2. Sensitivity of IGF1R and CTLA4 estimates to individual instruments



Each row omits the named SNP. Original reference frequencies; 25 correlated omission estimates. TSHR and BBJ CTLA4 are single-instrument analyses and cannot undergo omission.